

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

C05:

Measure network performance using key metrics with simulation tools
(BL4,BL5)

Assessment Tool Weightage: 40%

QUESTIONS: 3

Total Marks:30

Annexure A

Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability
2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.
3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Understanding of Problem & Constraints	20%	Clearly identifies all constraints and fully understands the problem requirements	Identifies most constraints with minor gaps in understanding	Identifies some constraints but misses key aspects	Shows little or no understanding of constraints or problem
Problem Decomposition	15%	Breaks problem into logical, well-structured sub-problems effectively	Breaks problem into sub-parts with some logical flow	Attempts decomposition but lacks clarity or structure	No clear decomposition of the problem
Application of Constraints in Solution	20%	Applies all constraints correctly and consistently in solution design	Applies most constraints correctly with minor errors	Applies some constraints but with noticeable errors	Fails to apply constraints appropriately
Logical Reasoning & Strategy	15%	Uses strong, clear, and efficient reasoning strategies	Uses generally sound reasoning with minor gaps	Reasoning is partially correct but inconsistent	Reasoning is unclear or incorrect
Solution Accuracy & Feasibility	15%	Solution is fully correct, feasible, and optimized	Solution is mostly correct and feasible	Solution has errors or is partially feasible	Solution is incorrect or not feasible
Creativity & Innovation	5%	Demonstrates highly innovative and unique approach	Shows some creativity in approach	Limited creativity shown	No creativity; relies on basic or copied ideas
Clarity of Presentation	10%	Solution is clearly presented with excellent organization and explanation	Presentation is clear with minor issues	Presentation lacks clarity or organization	Poorly presented and difficult to understand

1. Rural Branch Office – Internet Connectivity and Security

Problem Analysis:

The branch is located in a rural area where high-speed wired broadband is unavailable. However, it needs cloud applications, video conferencing, and secure communication with headquarters. The main constraints are limited budget, reliability, bandwidth limitations, security, and future scalability.

Recommended Solution:

1. Internet Connection

- Use **4G/5G wireless broadband** as the primary connection if mobile coverage is available.
- If mobile connectivity is unreliable, use **satellite internet** as an alternative.
- For uninterrupted operation, use a **dual-WAN router** with automatic failover between two connections.

2. Networking Devices

- **Dual-WAN router:** Connects multiple internet links and provides automatic failover.
- **Managed Gigabit switch:** Connects computers, printers, IP phones, and other devices.
- **Wireless access points:** Provide Wi-Fi connectivity within the office.
- **VPN gateway:** Creates a secure connection with headquarters.
- **Firewall:** Controls incoming and outgoing network traffic.

3. Security Mechanisms

- Establish a **site-to-site VPN** between the branch and headquarters.
- Use **firewall rules** to block unauthorized traffic.
- Use **WPA3** for wireless security.
- Implement **strong passwords and multi-factor authentication (MFA)**.
- Keep routers, firewalls, and endpoints updated.

4. Bandwidth Management

- Apply **QoS (Quality of Service)** to prioritize video conferencing and business-critical cloud applications.
- Limit unnecessary streaming and recreational traffic.
- Use bandwidth monitoring to identify congestion.

5. Reliability and Scalability

- Dual-WAN failover ensures that business operations continue if the primary connection fails.
- Start with a low-cost 4G/5G connection and upgrade to higher bandwidth when the branch grows.
- The managed switch and scalable router can support additional users and devices.

Justification:

This solution provides a balance between **cost, reliability, performance, and security**. 4G/5G avoids expensive wired infrastructure, while dual-WAN provides backup connectivity. QoS ensures that important applications receive sufficient bandwidth, and VPN/firewall protection secures communication with headquarters.

2. University Campus – Wireless Network for 3,000+ Students

Problem Analysis:

The university requires campus-wide wireless connectivity for more than 3,000 simultaneous users. The major constraints are high user density, limited access points, interference, security, coverage, performance, and budget.

Proposed Wireless Network Design:

1. Access Point Placement

- Conduct a **wireless site survey** before installation.
- Place APs in high-density areas such as:
 - Classrooms
 - Laboratories
 - Libraries
 - Hostels
 - Common areas
- Avoid placing APs behind thick walls or near sources of electromagnetic interference.
- Use directional antennas where appropriate in long corridors or specific high-density areas.

2. Frequency Band Selection

- Prefer **5 GHz** for high-speed and high-density areas because it provides more non-overlapping channels and less interference than 2.4 GHz.
- Use **2.4 GHz** mainly for devices that require greater range or support only 2.4 GHz.
- Where supported, use **6 GHz/Wi-Fi 6E** for newer devices.
- Proper channel planning should be used to prevent neighboring APs from interfering with each other.

3. High-Density Configuration

- Use enterprise-grade **Wi-Fi 6/6E access points** capable of handling many simultaneous users.
- Configure appropriate transmit power rather than simply increasing it.
- Use band steering to encourage compatible devices toward higher-capacity bands.
- Enable load balancing between APs.

4. Authentication and Security

- Use **WPA3-Enterprise** with **802.1X/RADIUS authentication**.
- Each student should authenticate using their university credentials.
- Create separate networks/VLANs for:
 - Students
 - Faculty
 - Administration
 - Guests
 - IoT devices

5. Cost Management

- Instead of installing APs everywhere, prioritize **high-density locations**.
- Use centralized wireless management to monitor AP performance.
- Expand AP deployment gradually based on usage statistics.

Justification:

Using Wi-Fi 6/6E, proper AP placement, 5 GHz/6 GHz capacity, channel planning, and load balancing improves performance for thousands of users. WPA3-Enterprise and 802.1X provide strong authentication. VLANs isolate different user groups and improve security.

3. Financial Institution – Secure Internal Network

Problem Analysis:

A financial institution handles highly sensitive customer information, so the network must provide strong confidentiality, integrity, availability, compliance, and access control without significantly increasing cost or reducing performance.

Proposed Secure Network Architecture:

1. Network Segmentation

Divide the network into separate **VLANs/security zones**:

- **Employee VLAN:** Normal employee systems.
- **Server VLAN:** Application and database servers.
- **Customer/Online Banking DMZ:** Public-facing banking services.
- **Management VLAN:** Network and security administration.
- **Guest VLAN:** Internet-only guest access.

Sensitive databases should not be directly accessible from employee or guest networks.

2. Firewall Deployment

Use **next-generation firewalls (NGFWs)** at important network boundaries.

A simplified architecture is:

Internet → Firewall → DMZ → Internal Firewall → Internal Network

Firewall rules should follow the **least-privilege principle**, allowing only required traffic.

3. Authentication

- Use **MFA** for employees accessing sensitive systems.
- Use **802.1X** for network access control.
- Use **role-based access control (RBAC)** so employees can access only the information required for their jobs.
- Use strong password policies and centralized identity management.

4. Intrusion Detection and Prevention

Deploy **IDS/IPS** to identify:

- Unauthorized access
- Malware
- Port scanning
- Suspicious network traffic
- Possible attacks

Security logs should be centrally monitored using a **SIEM** system.

5. Performance Optimization

- Use VLANs to reduce unnecessary broadcast traffic.
- Place security controls at appropriate boundaries rather than inspecting every type of traffic unnecessarily.
- Use hardware-accelerated firewalls where required.
- Apply QoS to business-critical applications.

6. Cost and Compliance

- Use centralized security management to reduce operational complexity.
- Prioritize strong controls around customer-data systems.
- Maintain logs for auditing and compliance.
- Regularly update firewall, IDS/IPS, and endpoint security systems.

Justification:

Network segmentation prevents an attacker who compromises one system from easily reaching sensitive databases. Firewalls control communication between security zones, while MFA and RBAC prevent unauthorized users from accessing sensitive information. IDS/IPS provides continuous monitoring and early detection of attacks.